Nikolaos Goutzoulias

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Education

Columbia University Aug 2024 - Dec 2025
Computer Science MS New York, NY

- GPA: 3.54/4.00
- Relevant Courses: Unsupervised Learning, Advanced Deep Learning, Advanced Big Data and AI

National Technical University of Athens Jul 2024
Electrical and Computer Engineering - Integrated MEng Athens, GR

- Grade: 8.7/10 (top 5% of 400 students)
- Relevant Courses: Deep Learning, Databases, Advanced Algorithms

Professional Experience

Ripple May 2025 - Aug 2025
Blockchain Data Scientist Intern New York, NY

- Productionized a **Databricks ETL pipeline using SQL** to calculate DeFi metrics from **raw, on-chain XRPL DEX data**. The analysis **helped the product team** measure the ecosystem's health and was used by leadership.
- Built **LangGraph-style multi-agent pipelines (Gemini 2.5 + n8n)** orchestrated with Kubernetes, encoding task memory, adaptive orchestration to autonomously analyze and write smart contracts, with task success of 95%.
- Developed a **reusable data pipeline** using **Python, PySpark, and rigorous A/B Testing** to automate the calculation of rolling asset balances, eliminating 100s of spreadsheets. Deployed as FastAPI microservices on AWS for Treasury use.

Columbia University Jan 2025 - May 2025
Course Assistant - Analysis of Algorithms New York, NY

- Guided 150+ students in mastering data structures, **dynamic and linear programming**, and **nonlinear optimization** techniques, created 10+ problem walkthroughs and a DP cheat-sheet adopted by the course.

Graduate Research Assistant – Software Systems Lab Sep 2024 - Apr 2025

- Collaborated with Meta researchers through **Git** to create a **browser API** in **C++** for Chrome by applying REST contracts and measuring user engagement without cross-site tracking.

EmDoT S.A. Oct 2023 - Jul 2024
Machine Learning Engineer - Thesis Collaboration Athens, GR

- Modeled **customer behavior** through deep learning models with **PyTorch** using **CUDA**, and a random forest with **Sklearn**, boosting prediction accuracy by 15% for resource allocation KPIs in an insurance company.
- Optimized model performance by 20% and accuracy by performing **feature engineering**, incorporating parameters such as time of day and **drift-aware metrics**. Built **monitoring dashboards in Tableau** to increase usability.
- Engaged with cross-functional teams, directly **interacted with customers** and **deployed ML models** in industrial use.

Data Science Intern Jul 2023 - Sep 2023

- Delivered algorithms for smoothing Bin Fill Level data, using **SQL**, **Tableau**, and statistical models like **Prophet and SARIMAX** to predict the trend of the data. Reduced noise in data by 20% while maintaining 95% of original signal.

Projects

NutritionAI: LLM Nutritional Personalized Assistant Jan 2025 - Mar 2025
Columbia University

- Built and deployed NutritionAI, a personalized LLM nutrition assistant using **fine-tuned Llama3 8B with LoRA + RAG**, exposed via **FastAPI microservices** for real-time user queries.

Evaluating LLM Resilience to Adversarial Data Poisoning Sep 2024 - Dec 2024
Columbia University

- Designed a method to assess the resilience of **RoBERTa LLM** in sentiment analysis against **adversarial data poisoning**, which revealed a model accuracy drop of up to 15%.

Skills

Programming Languages: **Python**, **C++**, TypeScript, Matlab, **SQL**, R

ML Frameworks: **Tensorflow**, CUDA, **PyTorch**, Keras, **Scikit-learn**, Transformers, XGBoost

Cloud, Development and Other Tools: **GCP**, Databricks, Docker, Kubernetes, PowerBI, **PySpark**, API design, **Matplotlib**, **GitHub**, Tableau, FastAPI, LoRA/QLoRA, RLHF/DPO, LangGraph, Solidity